

CAPM are strongest in stocks that have small market capitalizations and trade in illiquid markets where information is not promulgated efficiently.

In summary, we expect that when the assumptions behind the CAPM are violated, additional risk premiums should manifest themselves. These include macro factors, which should affect investors' nonfinancial considerations, effects associated with the asymmetric treatment of risk, illiquidity and transactions costs, and taxes. We should expect failures of the CAPM to be most apparent in illiquid, inefficient markets. The assumption, in particular, of perfect information is one of the reasons why modern economists no longer believe that markets are efficient in the form the original CAPM specified.

6. The Fall of Efficient Market Theory

Today, economists do not believe in perfectly efficient markets.¹³ In fact, markets cannot be efficient in their pure form. The modern notion of market near-efficiency is developed by Sanford Grossman and Joseph Stiglitz (1980), which forms part of the collection of papers for which Stiglitz was awarded his Nobel Prize in 2001. Grossman and Stiglitz describe a world in which markets are nearly efficient, and in doing so they address a conundrum that arises from the costless information assumption of the CAPM. Suppose that it is costly to collect information and to trade on that information, as it is in the real world. Then, if all information is in the price already, why would anyone ever invest in gathering the information? But if no one invests in gathering the information, how can information be reflected in security prices so that markets are efficient? It is then impossible that markets be efficient in their pure form.

Grossman and Stiglitz develop a model in which markets are near-efficient. Active managers search for pockets of inefficiency, and in doing so cause the market to be almost efficient. In these pockets of inefficiency, active managers earn excess returns as a reward for gathering and acting on costly information. In the assumptions of the CAPM discussed above, we should expect these pockets of inefficiency to lie in market segments that are illiquid, with poor information dissemination and where outsized profits may be hard to collect because trading on these anomalies will likely move prices. Whether active managers actually do earn excess returns for their investors (rather than for themselves) is a topic that we cover extensively in the third part of this book.

¹³The "classical" notions of weak, semi-strong, and strong efficiency were laid out by Fama (1970) and are obsolete. Fama was awarded the Nobel Prize in 2013. In that year, the Nobel Prize committee also gave Robert Shiller the prize, representing the opposite viewpoint of behavioral, or non-rational, influences on financial markets.